

## EE 527: Analysis and Design of Power Electronic Converters

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| <b>Course Code</b>   | EE 527                                                                                                              |
| <b>Course Name</b>   | Analysis and Design of Power Electronic Converters                                                                  |
| <b>L-T-P-C</b>       | 3-0-0-3                                                                                                             |
| <b>Prerequisites</b> | EE 309 - Power Electronics or Equivalent or instructor Consent                                                      |
| <b>Intended for</b>  | BTech Final year/M.Tech./MS/PhD                                                                                     |
| <b>Distribution</b>  | Core course for M.Tech. Power Electronics and Drives (PED), Elective for final year Electrical Engineering, MS. PhD |
| <b>Approval</b>      | 17th BoA (May 2017); 13th Senate (Jun 2017)                                                                         |

### Course Contents

- **Introduction:** [2 Lectures]
  - Review of basic power electronics concepts, power control through switching overview of power devices, converters and emerging applications
- **AC-DC Converters:** [11 Lectures]
  - Introduction types of AC-DC converters such as line commutated type, Multipulse topologies, and PWM rectifiers
  - Power quality issues related to AC-DC converters, Mitigation through AC-DC power factor correction circuits in single phase and three phase application and through multi-level AC-DC conversion
  - Modelling, design and control of power factor correction circuits
  - Bi-directional PWM rectifiers: Dynamic switching function models, control and applications as front end converters
- **DC-AC converters:** [12 Lectures]
  - Introduction to DC-AC converters including multilevel, inverters for open ended load configurations and their switching strategies
  - Voltage source and current source converters - topologies and PWM techniques
  - Single phase and three phase inverters: Dynamic models, control methods and applications as grid connected converter
  - Applications in low frequency AC synthesis and three-phase PWM techniques
  - Calculation of switching and conduction losses in DC-AC converters
- **AC-AC Converters:** [10 Lectures]
  - Introduction to AC-AC converters including matrix converters and multi-stage converters
  - Single phase and three phase matrix converters - topologies and PWM techniques
  - Modelling and control strategies of matrix converters
  - Multi-stage converters: Voltage link & current link topologies, dynamic models, Control methods and applications
- **High frequency link Converters:** [7 Lectures]
  - Introduction to high frequency link converters including resonant inverters and high frequency rectifiers
  - Basic concepts on inverters with resonant DC link, high-frequency rectifiers, and de-de resonant converters and their applications in energy storage system

### Text Books

1. Mohan N., Undeland T.M. and Robbins W.P., **Power Electronics -Converters, Applications and Design**, 3rd Edition, Wiley India. 2008
2. L. Umanand, **Power Electronics: Essentials & Applications**, Wiley India, 2009.

### References

1. R.W. Erickson, D. Maksimovic, **Fundamentals of Power Electronics**, 2nd Edition, Kluwer Academic Publishers.
2. Bin Wu, **High-Power Converters and AC Drives**, IEEE Press, John Wiley & Sons, 2006.
3. Rashid M., **Power Electronics- Circuits, Devices and Applications**, 3rd Edition, Pearson Education.
4. A. LPressman, **Switch Mode Power Supply Design**, McGraw-Hill, 1999.
5. R.S. Ramshaw, **Power Electronics Semiconductor Switches**, Chapman & Hall, 1993.
6. D. Grahamc Holmes, Thomas A. Lipo, **Pulse width modulation for power converters: principles and practice**, John Wiley & Sons, 2003.